The Journal of Open Source Software

PyVWF: An open Python framework for bias-corrected wind power simulation from reanalysis data

Ellyess F. Benmoufok[1], **Simon C. Warder**[1], **and Matthew D. Piggott**[1]

1 Department of Earth Science and Engineering, Imperial College London, London SW7 2AZ, UK

DOI: 10.xxxxxx/draft

Software

- Review
- Repository
- Archive

Editor:

Submitted: 17 July 2026
Published: unpublished

Summary

`PyVWF` (the Python Virtual Wind Farm) is an open-source Python framework that converts atmospheric reanalysis data into bias-corrected wind power generation. It re-implements, in a modular and extensible Python codebase, the Virtual Wind Farm (VWF) methodology that underpins the wind simulations on Renewables.ninja (Pfenninger & Staffell, 2016; Staffell & Pfenninger, 2016). Given gridded reanalysis winds (ERA5 (Hersbach et al., 2020); the original VWF used MERRA-2 (Gelaro et al., 2017)), turbine metadata, smoothed power curves, and observed generation, `PyVWF` extrapolates wind to hub height and interpolates to turbine locations, learns per-cluster scale-and-offset corrections by comparing simulated and observed capacity factors, applies these corrections to the ERA5 wind speeds, and reconverts the corrected winds to capacity factor. Unlike API-only or general-purpose reanalysis-to-power tools, it exposes the full **training** workflow for the bias-correction factors and lets researchers compute them at finer spatial and temporal resolution than the conventional national-scale factors, through configurable spatial clustering and temporal grouping. The package implements the granular bias-correction method of Benmoufok et al. (2024). A redistributable open power-curve library (Duffy, 2020) ships with the package (see *Functionality*), so the workflow runs out of the box; users supply their own curve library for manufacturer-specific production work. Experimental gridded interpolation and machine-learning extensions are maintained on a separate development branch.

Statement of need

Continental-scale energy system models such as `PyPSA-Eur` (Hörsch et al., 2018) depend on wind power time series derived from reanalysis, but reanalysis-derived capacity factors can carry biases of up to ± 50 % (Staffell & Pfenninger, 2016). These biases can propagate through non-linear power conversion and spatial aggregation and, when employed within energy system models, lead to misleading conclusions about generation mix, transmission investment, and system cost. The original VWF model corrects this bias, but its correction is accessible only through the closed Renewables.ninja API and cannot be inspected or retrained. Existing *open* tools cover only part of the problem: general-purpose converters such as `atlite` (Hofmann et al., 2021) omit observation-based correction entirely, while validated global products such as ETHOS.RESKit (Peña-Sánchez et al., 2026) ship a fixed national-resolution calibration built on a single global wind-speed curve, not a retrainable pipeline. `PyVWF` fills this gap. It implements a peer-reviewed method (Benmoufok et al., 2024) that extends the validated VWF approach (Staffell & Pfenninger, 2016) to ERA5 and to sub-national, seasonal resolution, and it reports distributional diagnostics in addition to mean error. Corrections can be computed at whatever spatial and temporal granularity the observation record supports, then inspected and retrained as that record grows. Research extensions on a separate development branch interpolate corrections onto a regular grid and export them for use with `atlite`/`PyPSA-Eur` workflows, supporting resource assessment and sensitivity studies across scales. The framework is intended for researchers in energy systems and climate science, and for power system analysts, who need calibrated wind resource simulations whose correction step they can reproduce and examine.

Functionality

`PyVWF` supports a turbine-, regional-, or national-scale workflow, in which a *cluster* is a group of nearby turbines or grid points sharing one set of bias-correction parameters. Wind is extrapolated to hub height via a log wind profile using ERA5 surface roughness (or roughness derived from 10 m/100 m wind shear) and bilinearly interpolated to turbine locations. Per-cluster, per-time-slice scale-and-offset factors (`w_corrected = α·w + β`) are then learned by comparing simulated and observed capacity factors (Benmoufok et al., 2024), with user-configurable spatial clustering and temporal grouping (fixed, seasonal, bimonthly, monthly). Observed generation and site metadata are supplied through `ObservationSource` adapters resolved via a registry, so a new region is added by writing an adapter rather than editing the pipeline. Turbine-level adapters (code, not data) for Denmark, Germany, and the United Kingdom ship with the package; the observation datasets themselves are user-supplied. The `vwf.viz` module is designed so that a learned correction can be examined directly rather than applied unseen. It provides distributional diagnostics (capacity-factor histograms, empirical CDFs, and quantile-quantile plots), maps of the learned factors, and error-versus-cluster-count curves for hyperparameter selection.

A redistributable open power-curve library of 69 real machines and 7 composites from the NREL/turbine-models archive (Duffy, 2020) (BSD-3-Clause), smoothed by an independent reproduction of the VWF method (Staffell & Pfenninger, 2016), ships with the package, so an automated `pytest` suite runs the full workflow on synthetic data in under a minute with no reanalysis download. The granular bias-correction method has been applied in peer-reviewed studies of European and UK wind resources (Benmoufok et al., 2024; Wang et al., 2026).

State of the field

Open tools in this area include the general-purpose converter `atlite` (Hofmann et al., 2021), which applies no observation-based correction, and validated global products such as ETHOS.RESKit (Peña-Sánchez et al., 2026), which pairs a measurement-trained wind-speed correction with a national capacity-factor calibration; the VWF code behind Renewables.ninja (Pfenninger & Staffell, 2016; Staffell & Pfenninger, 2016) remains closed and largely dormant. For a ready-made global assessment, RESKit is the stronger choice. `PyVWF` is complementary. It implements the peer-reviewed granular bias-correction method of Benmoufok et al. (2024), which adapts the VWF approach of Staffell & Pfenninger (2016) to ERA5 and refines it to sub-national and seasonal resolution, and it exposes the training step of that correction as a pipeline the user can inspect and retrain; the calibrations above ship as fixed national products.

AI disclosure

AI-assisted tools were used to help with software refactoring, test scaffolding, and editing portions of this manuscript for clarity. All technical content, claims, methods, and citations were reviewed and verified by the authors, who take full responsibility for the final software and text.

Acknowledgements

The original VWF model was developed by Iain Staffell, and Renewables.ninja by Stefan Pfenninger and Iain Staffell. The authors thank collaborators on the underlying bias-correction

research (Benmoufok et al., 2024).

References

Benmoufok, E. F., Warder, S. C., Zhu, E., Bhaskaran, B., Staffell, I., & Piggott, M. D. (2024). Improving wind power modelling through granular spatial and temporal bias correction of reanalysis data. *Energy*. https://doi.org/10.1016/j.energy.2024.133759

Duffy, P. (2020). *Turbine-models*. National Renewable Energy Laboratory. https://doi.org/10.11578/dc.20210112.1

Gelaro, R., McCarty, W., Suárez, M. J., Todling, R., Molod, A., Takacs, L., Randles, C. A., Darmenov, A., Bosilovich, M. G., Reichle, R., Wargan, K., Coy, L., Cullather, R., Draper, C., Akella, S., Buchard, V., Conaty, A., Silva, A. M. da, Gu, W., … Zhao, B. (2017). The modern-era retrospective analysis for research and applications, version 2 (MERRA-2). *Journal of Climate*, *30*, 5419–5454. https://doi.org/10.1175/JCLI-D-16-0758.1

Hersbach, H., Bell, B., Berrisford, P., Hirahara, S., Horányi, A., Muñoz-Sabater, J., Nicolas, J., Peubey, C., Radu, R., Schepers, D., Simmons, A., Soci, C., Abdalla, S., Abellan, X., Balsamo, G., Bechtold, P., Biavati, G., Bidlot, J., Bonavita, M., … Thépaut, J. N. (2020). The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, *146*, 1999–2049. https://doi.org/10.1002/QJ.3803

Hofmann, F., Hampp, J., Neumann, F., Brown, T., & Hörsch, J. (2021). Atlite: A lightweight python package for calculating renewable power potentials and time series. *Journal of Open Source Software*, *6*(62), 3294. https://doi.org/10.21105/joss.03294

Hörsch, J., Hofmann, F., Schlachtberger, D., & Brown, T. (2018). PyPSA-Eur: An open optimisation model of the European transmission system. *Energy Strategy Reviews*, *22*, 207–215. https://doi.org/10.1016/j.esr.2018.08.012

Peña-Sánchez, E. U., Dunkel, P., Winkler, C., Heinrichs, H., Prinz, F., Weinand, J. M., Maier, R., Dickler, S., Chen, S., Gruber, K., Klütz, T., Linßen, J., & Stolten, D. (2026). Towards high resolution, validated and open global wind power assessments. *Nature Communications*, *17*, 539. https://doi.org/10.1038/s41467-026-68337-z

Pfenninger, S., & Staffell, I. (2016). Long-term patterns of european PV output using 30 years of validated hourly reanalysis and satellite data. *Energy*, *114*, 1251–1265. https://doi.org/10.1016/J.ENERGY.2016.08.060

Staffell, I., & Pfenninger, S. (2016). Using bias-corrected reanalysis to simulate current and future wind power output. *Energy*, *114*, 1224–1239. https://doi.org/10.1016/J.ENERGY.2016.08.068

Wang, Y., Warder, S. C., Benmoufok, E. F., Wynn, A., Buxton, O. R. H., Staffell, I., & Piggott, M. D. (2026). Geographic variability in reanalysis wind speed biases: A high-resolution bias correction approach for UK wind energy. *Energy Conversion and Management*. https://doi.org/10.1016/j.enconman.2026.121066